Daiwei Chen

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Research Focus: Generative Recommendation & Personalization

Ph.D. researcher working on generative retrieval, Semantic-ID representations, and learning individual preferences. First-author GTI (**EMNLP 2026**) and PAL (**ICLR 2025**), with LinkedIn GenAI research experience.

Education

University of Wisconsin-Madison — Madison, WI
Ph.D. in Electrical and Computer Engineering — *Sep 2023 – Present*
Advisor: Ramya Korlakai Vinayak

University of Pennsylvania — Philadelphia, PA
M.S. in Systems Engineering — *Aug 2021 – May 2023*

Zhejiang University — Hangzhou, China
B.S. in Psychology — *Sep 2017 – Jun 2021*

Industry Research Experience

LinkedIn — Mountain View, CA

Responsible AI - Research Intern — Jun 2026 – Sep 2026

- Developed **Multi-Teacher On-Policy Distillation (MOPD)** to address the safety–capability trade-off in LLM post-training.

Generative AI - Research Intern — May 2025 – Aug 2025

- Developed **GTI** for LLM-based candidate retrieval; achieved a **21.6% relative Precision@5 gain** over vanilla SFT in offline industrial evaluation (Good Match criterion).
- Introduced a grounding stage that learns new **Semantic-ID embeddings** from paired text while freezing the LM backbone, followed by fine-tuning for generative retrieval.

Selected Research & Publications

Grounded Token Initialization for New Vocabulary in LMs for Generative Recommendation
D. Chen, Z. Fu, C. Jiang, H. Zhang, R. Zhou, T. Wang, C. Yao, G. Li, R. Cai, Y. Cao, R. Jiang, F. Borisyuk, J. Shen, J. Wu, R. K. Vinayak
EMNLP 2026 *Main Conference*
Grounding Semantic-ID embeddings in pretrained LMs improves generative retrieval and recommendation.

LinkedOut: Linking World Knowledge Representation Out of Video LLM for Next-Generation Video Recommendation
H. Zhang, Y. Lu, L. Wang, Y. Li, D. Chen, Y. Xu, Y. Fu
Findings of CVPR 2026
Fusing world-knowledge representations from video LLMs enables personalized recommendation from raw frames.

PAL: Sample-Efficient Personalized Reward Modeling for Pluralistic Alignment
D. Chen, Y. Chen, A. Rege, Z. Wang, R. K. Vinayak
ICLR 2025 | *NeurIPS 2024 Workshop (**Spotlight**)* | *ICML 2024 Workshop (**Oral**)*
Shared preference prototypes enable sample-efficient personalization across text and image preference tasks.

One Model, Many Modes: Evolving System Prompt Ensembles for Creative Generation
D. Chen, A. He, B. Yao, J. Hu, R. K. Vinayak
Under review
Evolved system-prompt ensembles improve novelty and diversity without retraining or sacrificing quality.

Compositional Self-Improvement
C. Shin, D. Chen, J. Cooper, B. Lake, F. Sala, R. K. Vinayak
ICML 2026 *– 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents*
Composing reliable subproblem predictions enables self-training beyond the original supervision regime.

Service & Awards

Reviewer: NeurIPS 2026, ICLR 2026, CVPR 2026, EACL 2026, COLM 2026.
Awards: Lambda.ai Research Grant Sponsorship (2025); UPenn Outstanding Research Award (2023).